A MORPHISM IS FORMALLY UNRAMIFIED EXACTLY WHEN ITS SHEAF OF DIFFERENTIALS VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed. The author's slip 'defined by a an ideal' is reproduced verbatim.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Formally unramified morphism, tested against first order thickenings. Same chapter, Definition environment labelled definition-formally-unramified (source line 798, Stacks tag 02H8); invoked at proof line 934. Supplies the two-morphism lifting condition that the converse direction has to verify; the verification itself is retained in the counted span.
- The conormal sequence of the diagonal and the identification of the sheaf of differentials with J/J^2. Morphisms, Lemma labelled morphisms-lemma-differentials-affine (canonical Stacks tag 01UT); the author says at proof lines 903-905 that the diagrammatic set-up is recalled from its proof. Supplies the open W containing the image of the diagonal, the ideal J of the diagonal, the square-zero ideal J^2 and the identification Omega_{X/S} = J/J^2 used throughout.

Source reference. Original lemma 01UT: sources/stacks-chow/tags.txt, source lines 2422–2422. Cross-chapter standard prerequisite invoked at proof line 905 as the place where the diagonal set-up is recalled from; the author reproduces the needed facts (W, J, J^2, Omega_{X/S} = J/J^2) inside the counted span.

Source reference. Original definition 02H8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 798–798. Same-chapter pointer invoked at proof line 934 for the lifting condition that the converse direction verifies; that verification is retained verbatim in the counted span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Then f is formally unramified if and only if $\Omega_{X/S} = 0$.*

Proof. We recall some of the arguments of the proof of Morphisms, Lemma 01UT. Let $W \subset X \times_S X$ be an open such that $\Delta : X \to X \times_S X$ induces a closed immersion into W. Let $\mathcal{J} \subset \mathcal{O}_W$ be the ideal sheaf of this closed immersion. Let $X' \subset W$ be

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the closed subscheme defined by the quasi-coherent sheaf of ideals $\mathcal{J}^2$. Consider the two morphisms $p_1, p_2 : X' \to X$ induced by the two projections $X \times_S X \to X$. Note that p_1 and p_2 agree when composed with $\Delta : X \to X'$ and that $X \to X'$ is a closed immersion defined by a an ideal whose square is zero. Moreover there is a short exact sequence

$$0 \to \mathcal{J}/\mathcal{J}^2 \to \mathcal{O}_{X'} \to \mathcal{O}_X \to 0$$

and $\Omega_{X/S} = \mathcal{J}/\mathcal{J}^2$. Moreover, $\mathcal{J}/\mathcal{J}^2$ is generated by the local sections $p_1^\sharp(f) - p_2^\sharp(f)$ for f a local section of $\mathcal{O}_X$.

Suppose that $f : X \to S$ is formally unramified. By assumption this means that $p_1 = p_2$ when restricted to any affine open $T' \subset X'$. Hence $p_1 = p_2$. By what was said above we conclude that $\Omega_{X/S} = \mathcal{J}/\mathcal{J}^2 = 0$.

Conversely, suppose that $\Omega_{X/S} = 0$. Then $X' = X$. Take any pair of morphisms $f_1', f_2' : T' \to X$ fitting as dotted arrows in the diagram of Definition 02H8. This gives a morphism $(f_1', f_2') : T' \to X \times_S X$. Since $f_1'|_T = f_2'|_T$ and $|T| = |T'|$ we see that the image of T' under (f_1', f_2') is contained in the open W chosen above. Since $(f_1', f_2')(T) \subset \Delta(X)$ and since T is defined by an ideal of square zero in T' we see that (f_1', f_2') factors through X'. As $X' = X$ we conclude $f_1' = f_2'$ as desired. $\square$